RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Thursday, 8 October 2026, 10:53 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:55 AM
Time taken	1 min 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int upper_bound(int tails[], int size, int target) {
    int low = 0, high = size;
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (tails[mid] <= target) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int longestNonDecreasingSubsequence(int seq[], int n) {
    if (n <= 0) return 0;

    int tails[n];
    int len = 0;

    for (int i = 0; i < n; i++) {
        int idx = upper_bound(tails, len, seq[i]);

        if (idx == len) {
            tails[len] = seq[i];
            len++;
        } else {
            tails[idx] = seq[i];
        }
    }

    return len;
}

int main() {
    int sequence[] = {-1, 3, 4, 5, 2, 2, 2, 2, 3};
    int n = sizeof(sequence) / sizeof(sequence[0]);

    int result = longestNonDecreasingSubsequence(sequence, n);

    printf("%d\n", result);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔

	Input	Expected	Got	
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course